

# SIMRAN MOHAPATRA

Cloud Engineer | Azure & AWS | MEng, University of Maryland  
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## PROFILE

Cloud Engineering graduate student at the University of Maryland with over three years of enterprise production experience at DXC Technology — 4 Azure subscriptions, 150–200 Linux/Windows VMs, end-to-end P1/P2 incident ownership, and direct collaboration with Microsoft Tier-3 engineers. Evidence-driven approach to diagnostics under high-volume incident conditions, with hands-on depth across operations, monitoring, Terraform, GitHub Actions CI/CD, virtualization, and AI-cloud integration using the Claude API tool-use pattern. Seven cloud certifications across Microsoft, AWS, and HashiCorp.

## EDUCATION

### University of Maryland, College Park

Master of Engineering, Cloud Engineering | GPA: 3.72/4.00

Aug 2025 – Dec 2026

College Park, MD

Completed: Cloud Computing, Virtualization & Container Technologies, Linux System Administration, Networks & Protocols, Python, Data Storage & Databases, Software Engineering.

Fall 2026: Cloud Security, Information Security Tools, Advanced TCP/IP Networking.

### Silicon University

Bachelor of Technology, Computer Science Engineering | CGPA: 7.4/10

2018 – 2022

India

## PROFESSIONAL EXPERIENCE

### Cloud Support Engineer (Analyst 2)

DXC Technology — Enterprise Azure Managed Services

Nov 2023 – Jul 2025

India

- Administered 4 Azure production subscriptions with 150–200 mixed Linux/Windows VMs — compute, networking (VNets, NSGs, VPN), identity (RBAC), and monitoring (Azure Monitor, ARM templates) across dev, staging, and production.
- Owned P1/P2 incident response end-to-end via ServiceNow: triage, root-cause analysis, stakeholder communication, resolution, and post-mortem documentation for outages affecting 150+ servers.
- Diagnosed a recurring bulk VM outage (50–60 VMs, 200+ simultaneous tickets) — disproved the team's network-drop hypothesis by noticing all affected VMs belonged to the same resource group; escalated to Microsoft Tier-3 with telemetry, identifying a software-state issue on a Top-of-Rack switch; environment stabilized within 2 days.
- Led CrowdStrike global outage recovery (July 2024) — coordinated fleet-wide Windows VM recovery using Microsoft-issued ISO procedure; restored full service while incident bridges were still active.
- Led disaster recovery exercises with 100% service restoration and zero client-facing downtime — recognized with DXC Q3 Winner FY24 and DXC Recognition Grow 2024 awards.
- Automated VM provisioning, patch validation, and health checks via PowerShell, reclaiming ~5 hrs/week; authored operational runbooks reducing MTTR by ~40%; served as designated KT lead onboarding new engineers.

### Associate Professional Software Engineer

DXC Technology

May 2022 – Nov 2023

India (Remote)

- Resolved 150+ production incidents using structured ITSM workflows; analyzed logs and monitoring signals to identify root causes of performance and availability issues; contributed to process improvements reducing recurring incident volume by ~20%.

## PROJECTS

### FloodGate — Message-Driven EKS Autoscaling System

AWS EKS · SNS · SQS · KEDA · CloudWatch · IAM · Helm · eksctl · kubectl

2026

- Designed event-driven pipeline (SNS → SQS → EKS workers); configured KEDA ScaledObject to auto-scale pods 1→10 on queue depth with 15-second poll and 60-second cooldown; provisioned via eksctl, deployed KEDA via Helm, authored scoped IAM inline policies; validated under 100-message synthetic load via CloudWatch Container Insights.

### EventFlow — Microservices Campus Events Platform (Group)

EKS v1.31 · Docker · ECR · ALB · RDS PostgreSQL Multi-AZ · Terraform · GitHub Actions · Prometheus · Grafana · AWS WAF · KMS · Karpenter · React · Node.js · Python/FastAPI

Fall 2025

- Co-built production-grade 3-microservice platform on EKS with Karpenter autoscaling, Multi-AZ VPC (public/private/database subnet tiers), ALB path-based ingress, and Multi-AZ RDS PostgreSQL with KMS encryption at rest.

- Implemented GitHub Actions CI/CD (Trivy CVE scan → Docker build → ECR push → EKS rolling deploy); configured Prometheus + Grafana observability and Kubernetes Network Policies enforcing Zero Trust pod-level isolation.

#### **CampusCopilot — AI-Powered Student Assistant**

*Hackathon 2026*

React · Claude API · Tool-Use / Function Calling · Streaming UI

- Integrated Claude API with real-time tool-use (function calling) — the same agentic pattern used in production AIOps; designed 5 custom tool schemas wired into a live streaming response loop with a multi-panel React UI for expense tracking, budget management, and event recommendations.

#### **StackForge — 3-Tier AWS Infrastructure (Terraform)**

*2025*

Terraform · EC2 · RDS · VPC · ALB · Auto Scaling · CloudWatch · SNS

- Built modular Terraform project (networking, security, compute, storage, monitoring modules) deploying ALB → ASG web tier → RDS in private subnets with NAT Gateway; conditional count pattern for environment-specific resource toggling.

#### **TicketGuard — Multi-Service Lottery Verification Platform (Group)**

*Spring 2026*

AWS ECS Fargate · ALB · RDS PostgreSQL · Terraform (8 modules) · CloudWatch · CloudTrail · VPC Flow Logs · SNS · Secrets Manager · ACM · ECR · Flask

- Logging, Monitoring & Documentation Lead — built the full observability layer for a 2-service ECS Fargate platform: CloudWatch Logs, VPC Flow Logs, CloudTrail with S3 delivery, and ALB access logs.
- Authored 9 CloudWatch alarms (service CPU/memory, ALB 5XX, unhealthy targets, RDS CPU, login-failure spikes) wired to SNS; built a CloudWatch dashboard and metric filters for security events; authored runbook, troubleshooting guide, and architecture diagram.

#### **ClinicalGraph — Polyglot Healthcare Database System (Group)**

*Spring 2026*

PostgreSQL · MongoDB · Neo4j · Python · Docker Compose

- Designed PostgreSQL schema (constraints, indexes, triggers) and seed data for a healthcare transactional system; authored Neo4j Cypher queries for a medical knowledge graph where graph traversal gates PostgreSQL writes as a drug-interaction safety check.

### **TECHNICAL SKILLS**

**Cloud Platforms:** Microsoft Azure (IaaS, VNet, NSG, RBAC, Monitor, IoT Hub, ARM) • AWS (EC2, EKS, ECS Fargate, EFS, SNS, SQS, Lambda, IAM, CloudWatch, CloudTrail, ALB, ASG, S3, ECR, RDS, KMS, WAF, ACM, Secrets Manager)

**IaC & CI/CD:** Terraform (multi-module, state) • AWS CloudFormation • ARM Templates • GitHub Actions • Kustomize

**Containers, Orchestration & Virtualization:** Docker • Kubernetes (EKS) • KEDA • kubectl • eksctl • VMware Workstation

**Observability:** Prometheus • Grafana • Alertmanager • CloudWatch (Dashboards, Alarms, Logs, Metric Filters) • CloudTrail • VPC Flow Logs

**Scripting & Programming:** PowerShell • Python • Bash • Linux

**Databases & Tools:** PostgreSQL • MongoDB • Neo4j (Cypher) • SQLite • ServiceNow • Confluence • Git/GitHub

### **CERTIFICATIONS**

**Microsoft Azure:** AZ-104 (Administrator), AZ-900 (Fundamentals), DP-900 (Data Fundamentals)

**Amazon Web Services:** Solutions Architect – Associate, AI Practitioner (AIF-C01), Cloud Practitioner

**HashiCorp:** Terraform Associate

### **LEADERSHIP & PROGRAM EXPERIENCE**

#### **Desk Assistant — Conferences & Visitors Services, University of Maryland**

*2026 – Present*

- Support campus conference and visitor operations through front-desk service, information routing, event support, and issue resolution.

#### **Student Ambassador — DXC Technology × Silicon University**

*Jan – May 2022*

- Primary liaison between DXC corporate representatives and university faculty, staff, and graduating students; coordinated onboarding sessions, announcements, and formal corporate-academic communications.

#### **Core Team Member & Lead Organiser — ZEAL Community Service Club**

*Aug 2018 – May 2022*

- Co-organised the university's largest inclusive sports event for specially-abled children — 5 partner schools and orphanages, official tie-up with the Odisha state government; one of three lead organisers across scheduling, logistics, registration, fundraising, and budgets over 4 years.